



Society of Petroleum Engineers

SPE-229872-MS

Successful Integration of Routing and Scheduling Operations Research AI Models into Danish North Sea Helicopter Logistics

R. Dubourg and R. Lhadj Mohand, TotalEnergies EP Danmark, Esbjerg, Denmark; S. Perrier, TotalEnergies, Paris, France; A. -L. Debar and S. Pandey, TotalEnergies EP Danmark, Esbjerg, Denmark

Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229872-MS](https://doi.org/10.2118/229872-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

This paper was selected for presentation by an SPE program committee following review of information contained in an abstract submitted by the author(s). Contents of the paper have not been reviewed by the Society of Petroleum Engineers and are subject to correction by the author(s). The material does not necessarily reflect any position of the Society of Petroleum Engineers, its officers, or members. Electronic reproduction, distribution, or storage of any part of this paper without the written consent of the Society of Petroleum Engineers is prohibited. Permission to reproduce in print is restricted to an abstract of not more than 300 words; illustrations may not be copied. The abstract must contain conspicuous acknowledgment of SPE copyright.

Abstract

Enhancing logistics performance in the E&P industry is historically a major factor in cost efficiency. This paper presents an AI-driven operational tool supporting helicopter operations in the Danish North Sea, joining cost and carbon footprint benefits by enabling reductions of flight hours and rotations through continuous re-optimization of the seat and routes offerings. The primary challenge of such automated optimization solutions lies in integrating mathematically optimal outcomes with real-world operations that demand considerable flexibility.

This paper details the implementation and insights gained from an operations research-based (OR) optimization digital solution, internally named ‘PAX’. The tool provides operational recommendations by utilizing state-of-the-art models while considering real-life constraints. Method wise, the strategy is to integrate Operations Research algorithms open-sourced by Google AI, encompassing the Capacitated Vehicle Routing Problem with Time Windows and Pickup and Delivery model (CVRPTWPD) with business specific algorithmic approaches. The business specific challenges that need to be addressed include the implementation of an optimal split-delivery search component, the management of various time-window constraints, and finally the design of seats assignments and departures.

In the Danish part of the North Sea, where many near-neighbor platforms constitute a complex logistics network, the solution reaches its goal: scheduling and routings of helicopters are optimized, with a good compromise with operational flexibility. While operations research models are not frequently used in practice due to difficulties dealing with actual real-world processes, PAX is the result of a multi-disciplinary approach that succeeds in delivering prescriptive, optimal and operationally relevant recommendations. Since July 2024, the optimization of helicopters routing and scheduling and associated split delivery have successfully guided the operations teams to save multiple flights and run consistently the shortest combined routes. Importantly as well, the solution's design allows transparency and explainability. The optimization of operations is measurable, providing valuable insights: cost indicators and seat utilization are fully contextualized, enabling a complete understanding of logistics performance.

The main two novel components of this innovative method are:

1. the implementation of split delivery added to the classic VRP ([Vehicle Routing Problem](#)) models. Indeed, helicopters have the specificity to be generally too small vehicles for the statistical quantities to transport, which requires a powerful split delivery heuristic search method to ensure optimality of the VRP.
2. the multi-time window scenario approach, that enables the integration of a Passenger-Transportation-System with advanced open-source Operations Research models.

The shared lessons learned have potential to be applied in other contexts of our industry.

Introduction

TotalEnergies Exploration & Production Danmark (TEPDK) has, by design, large helicopter operations: typically 30 flights and 60 offshore landings per week, for an average of 180 passengers per day. The geography of the fixed offshore platforms with helideck is represented in [Fig. 1](#). Opportunities for synergies in flight routing and scheduling are numerous.

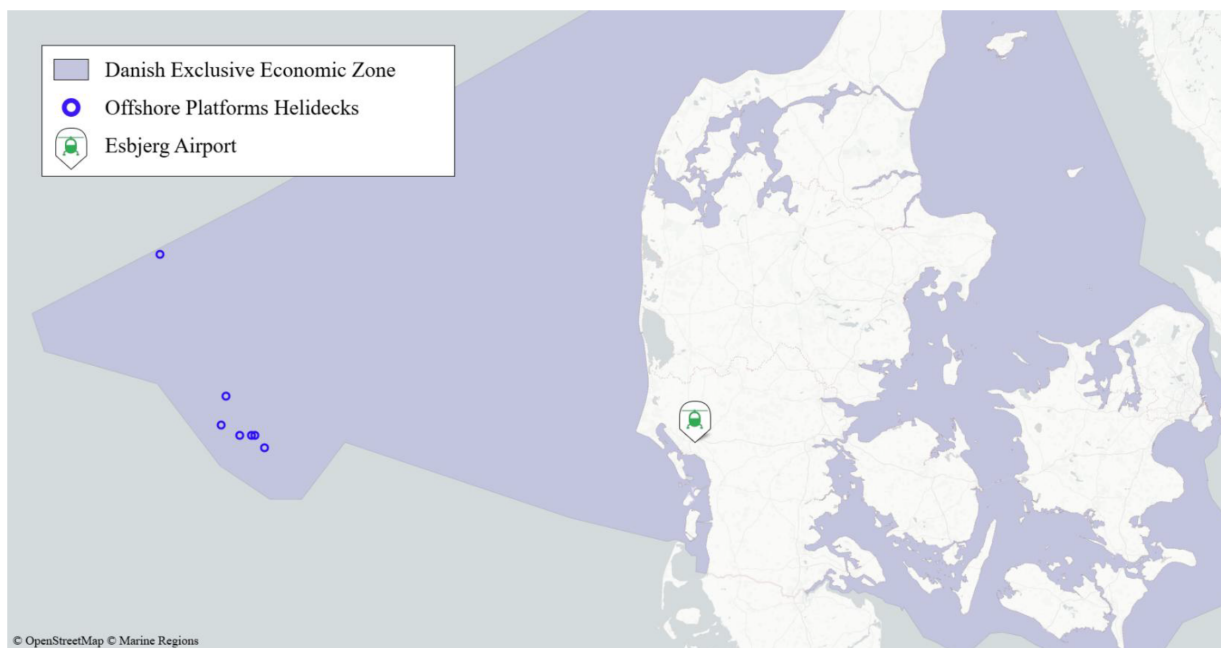


Figure 1—Fix-position nodes of the routing network in the Danish exclusive economic zone (EEZ). The nodes are the offshore platforms' helidecks and the so-called 'depot' node, Esbjerg airport, where all the helicopters start and end their flights. Drilling rigs and accommodation flotels with no fixed location are not on the map.

The solution, described in this article and named PAX, became TEPDK aviation operations team's decision-support, with seamless integration of automated optimization to the logistics workflow. It makes actionable recommendations on the opportunities to reduce the number of flights and to redefine the routes, to save an estimated target of 1MUSD per year, equivalent to saving one flight per week.

Designed around Google AI Operations Research open-source routing framework ([Google OR-Tools](#)), it combines four customized optimization models, from the reservations' inputs to a realistic and least disruptive optimized planning. Our solution uses PuLP library's solvers on top of the Google ones ([PuLP](#), [CP-SAT](#)).

Using its backbone algorithm, PAX is testing a new frontier: flagging 'high-cost seats' for operations teams to challenge or reconfirm, as these often require significant extra resources—such as additional flights—which, if avoided, can save money, reduce CO₂ emissions, and improve safety.

Our solution has direct and analog potential in other place performing similar operations, leveraging the virtuous circle: logistics excellence, cost, CO₂ minimization and safety.

Methods

In terms of technology, our solution falls into the category of Operations Research (OR), an ensemble of highly generalizable and customizable management and decision-making tools. They are widely used across all sectors and especially popular among logistics operations, telecom, and to a lesser extent in sectors such as finance, trading, healthcare, etc. Their integration into the industry remains challenging. There is a need to combine them with the human flexibility and reactivity sometimes required to deal with the realities of decision-making (Trefethen 1954).

In addition to the general practical complexity of operational routing and scheduling models, our helicopters' capacities are around 10 times smaller than the daily number of reservations, increasing computational complexity. Therefore, we decided to divide our work into four optimization models. Our models described in this section are at the intersection of integer and constraints programming. We used PuLP python library and Google CP-SAT solvers.

Input

Helicopter flight reservations and most operational constraints are automatically updated daily and serve as key data inputs. Some operational constraints are updated semiautomatically, such as the rigs' anchor age positions, or the weather impacts. The latter is integrated into the settings of helicopters' capacities (rarely) or the flight departure times.

Up to two weeks ahead, daily passenger reservations and daily planned helicopters schedules are fed to PAX every day.

Methodology

The overall methodology described in Fig. 2, is to break down the helicopters routing and scheduling optimization into smaller and more manageable tasks. Computation-wise, the main optimization brick is the OR-Tools framework for Capacitated Vehicle Routing Problem with Pickup and Delivery (CVRPTWPD). Before, the latter's input is the result of a split delivery optimization. After, its outputs are optimally post processed to reach the best trade off between a cost-efficient and least disruptive new planning of the helicopters' flights, with two optimization algorithms: the seats' allocations and the scheduling of helicopters' flights.

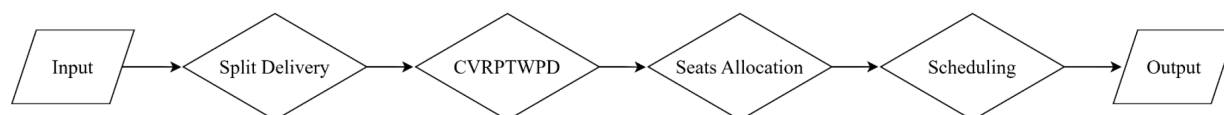


Figure 2—Programming flowchart of our method: a sequence of four optimization-based algorithms.

Split Delivery. In helicopter transportation scenarios, a unique challenge arises due to the typically small capacity of helicopters relative to the number of passengers to be transported. For instance, consider a group of 20 passengers traveling from an onshore airport (depot) to an offshore platform. If the helicopter has a capacity of 16 seats, the optimal solution is not necessarily to transport 16 passengers on the first flight and the remaining 4 on a second flight. A split such as 12 and 8 passengers might prove more efficient if the redistribution of seats enables a more optimal routing. This example highlights the concept of split delivery, which becomes particularly relevant when the group size exceeds the capacity of a single helicopter.

The reservations are grouped by start location, end location and flying time window, calculated using the Estimated Time of Departure (ETD) in the actual planning. In our case the outbound reservations, from

Esbjerg onshore airport to an offshore installation, and the return, inbound reservations, account for 99% of the passengers. The remaining transit reservations, transportation between offshore platforms, represent on average less than 2 passengers per day. Therefore, the transit reservations are split into unit groups of one passenger. The split delivery challenge remains for the outbound and inbound reservations.

The split delivery consists of dividing the groups of passengers with the same helicopters' reservations into smaller groups (Dror 1994). The split has several purposes: (i) ensuring that the groups do not exceed the helicopters capacities, (ii) decreasing the complexity of the general routing problem and (iii) guiding the routing solver towards preferable solutions with an optimized split demand input.

A hard and a soft constraint are used as levers, one after the other, to split the passengers into groups. First, the demands groups are distributed over allocated flying time windows (TW), then the groups are split to fit the helicopters capacities, with a simple modulo operation or via an optimization algorithm, (Archetti 2008). As the solution needs to anticipate for real world constraints, several levels of flexibility are tested:

- Constraining the passengers' flights within their original time window ("same_tw"),
- Releasing the constraint by allowing rescheduling of some passengers within the previous/next time window ("tw_plus_1", "tw_plus_minus_1"),
- Allowing rescheduling of some passengers within all airport working hours ("full_reshuffle", "full_reshuffle_pool"),
- Splitting the same reservations groups (all scenarios).

The Table 1 sums up the differences between the different scenarios, from most to least constrained around the planned reservations, leading to least to most disruptive optimized routing suggestions. The time windows and the splitting methods are decreasingly constraining the CVRPTWPD solver around the actual helicopter planning. It ensures that the scenarios will find the best trade off between the operability and the cost of the optimized routing suggestions.

Table 1—Preprocessing scenarios of the helicopters flights reservations. Time windows are the time intervals between the take off and the landing of a group of passengers. The splitting method is either calculated modulo the helicopters' capacities minimum or with an optimization model. The 5 best and distinct solutions from the latter model are being iterated over.

Scenarios	Time windows	Splitting method	Number of iterations
same_tw	Unchanged	Modulo helicopters capacities	1
tw_plus_1	+3 hours	Modulo helicopters capacities	1
tw_plus_minus_1	± 3 hours	Modulo helicopters capacities	1
full_reshuffle	[7:00, 20:00]	Modulo helicopters capacities	1
full_reshuffle_pool	[7:00, 20:00]	Optimization	1-5

Splitting the demand modulo the helicopters capacities means that we divide each reservation group by the smallest helicopter capacity into smaller groups. It is the easiest way to split the passengers, and it ensures that the actual reservations groups are preserved as much as possible.

Mathematical formulation for the optimization splitting method (for the "full_reshuffle_pool" scenario):

$$x_{p,f,i} = \begin{cases} 1 & \text{if passenger } p \text{ is assigned to flight } f \text{ and reservation leg type } i \text{ via installation } i \\ 0 & \text{otherwise} \end{cases}$$

Our problem is formulated as a multi-objective function with linear scalarization balancing flights and touchdowns minimization. A trade off between cost, safety and operability leads approximately to one flight saved being equivalent to 5 touchdowns saved. Thus, the linear scalarization coefficient α is set to 5.

$$\text{maximize} \sum_f (\alpha E_f + T_f), \quad \alpha = 5$$

$\sum_{f \in F} E_f$ counts the number of empty flights (potential savings), i.e. without assigned passenger reservations:

$$E_f = \begin{cases} 1 & \text{if } \forall (p, l, i) x_{p,f,l,i} = 0 \\ 0 & \text{otherwise} \end{cases}$$

T_f represents the number of non-visited platforms per flight, increasing with the potential touchdowns savings:

$$T_f = \sum_i \prod_{p,l} (1 - x_{p,f,l,i})$$

We also modeled realistic constraints to ensure operability of the solver's solution.

i. Only meeting the reservations D

$$\forall f, \quad (p, l, i) \notin D \Rightarrow x_{p,f,l,i} = 0$$

ii. One flight per reservation

$$\sum_{p,f,l,i} x_{p,f,l,i} = 1$$

iii. Helicopters flights seat capacities $L_{f,l}$ must not be exceeded

$$\forall (f, l), \quad \sum_{p,i} x_{p,f,l,i} \leq L_{f,l}$$

CVRPTWPD. Split Delivery Capacitated Vehicles Routing Problem with Time Windows and Pickups and Delivery (SD-CVRPTWPD) is the name of our optimization problem as commonly found in the literature (Ahkamiraad 2018). In other words, it is looking for the optimized routes of a fleet of limited capacity vehicles (here helicopters); to pick up and deliver passengers (or goods) within the time windows defined for each of them.

Google's OR-Tools, started in 2008 and made open-source in 2015, offers a flexible framework that can tackle this problem. On top of being open source, it is reliable, regularly upgraded and gathers a strong community of users helping each other. The routing solver has several advantages: it provides many first-solution heuristics, its guided local search improves the first solutions with metaheuristics, and their latest solver iterates to find an optimal solution and associated feasibility status.

Once the solver has found an optimal solution, the result is a route for each helicopter of the fleet, associated with its load in number of passengers and the time it reaches each stop along the way. For each helicopter, its route often contains more than one flight, i.e. more than one departure from the onshore airport. The helicopters' routes load is anonymized, and at this stage it misses who and when the people behind the reservations are flying. Several possibilities can arise, and therefore we decided to optimize the distribution of the passengers over the flights.

Seats Allocation. Once the solver has found an optimal solution, the result is a route for each helicopter of the fleet, associated with its load in number of passengers and the time it reaches each stop along the way. The demand matching is necessary for several reasons: it ensures that passengers get an assigned seat, it allows us to figure out if planned flights were kept unchanged and it ensures that a maximum number of people originally flying in the same flight f_1 are kept together in the same optimized flight f_2 . In this regard, we modelled mathematically each passengers' reservations as Boolean:

$$\forall f_1 \in F_1, \forall f_2 \in F_2, \quad \forall i \in f_1, \forall j \in f_2, \quad \forall p \in \{\text{passengers}\},$$

$$x_{p,i,j} = \begin{cases} 1 & \text{if the passenger } p \text{ in the planned flight reservation } i \text{ is moved to the optimized routing leg } j \\ 0 & \text{otherwise} \end{cases}$$

We added some constraints to ensure that every passenger is attributed to one and one seat only in the optimized flights. To finish, we defined an objective function, to match as much as possible the attributed seats in the optimized flights with the planned groups of passengers, via C_{f_1} and C_{f_2} respectively trying to group people with same planned and optimization scenario's flights.

$$\forall (m, n) \in \llbracket 1, 2 \rrbracket^2, m \neq n, \quad \forall f \in F_m \quad C_f = \max_{f' \in F_n} \left[\sum_p \sum_{i \in f} \sum_{j \in f'} x_{p,i,j} \right]$$

Objective:

$$\max \left[\sum_{f \in F_1 \cup F_2} C_f \right] \quad (F_1 \cap F_2 = \emptyset)$$

Scheduling. Once the optimal helicopter flight routes have been computed, the task is now to schedule the flights and to assign the right helicopter unit to a corresponding flight, considering the logistical constraints. These restrictions that capture the reality of the onshore and offshore operations are keys to ensure the operational feasibility of our solution.

The main objective is to assign an ETD and a helicopter unit to each of the previous optimized routes with the aim of minimizing the operational costs subject to a set of operational constraints. We consider the following binary decision variables:

$$x_{f,h,s} = \begin{cases} 1 & \text{if helicopter } h \text{ is assigned to flight } f \text{ at ETD } s \\ 0 & \text{otherwise} \end{cases}$$

The operational costs considered in the PAX calculations are: (i) the flying hour price (in DKK/hr) for each helicopter p_h and (ii) the fuel consumption (in DKK/hr) for each helicopter $p_{fuel} \times K_h$. By multiplying these quantities by the flight time T_f (in hour) we obtain the operational cost of flight f . This is reflected in the below objective function:

$$\min \sum_{f \in F} \sum_{h \in H_f} \sum_{s \in S} x_{f,h,s} \left(T_f (p_h + p_{fuel} K_h) + Id(s, S) \right)$$

The set F contains all the optimized routes that need to be performed, the set H_f contains all the helicopters that can perform flight f , based on seat capacity and the set S contains all the possible ETDs for a helicopter to take off from the airport, sorted from earliest to latest. Note that due to an operational constraint, it is more convenient to fly with the earliest ETDs instead of the latest. This is also reflected in the objective function as the minimization is also occurring on the index of the ETD chosen in set S . In fact, the term $Id(s, S)$ corresponds to the index of ETDs in set S .

This minimization of the objective function occurs subject to the following operational constraints:

- i. All flights must be performed:

$$\forall f \in F: \sum_{h \in H_f} \sum_{s \in S} x_{f,h,s} = 1$$

- ii. Maximum number of take-offs per ETD (e.g., for $N_s = 1$ we would have maximum 1 take-off per ETD):

$$\forall s \in S: \sum_{f \in F} \sum_{h \in H_f} x_{f,h,s} \leq N_s$$

- iii. Maximum number of flights per helicopter per day (e.g., for $N_h = 3$, we would have up to 3 flights per day for helicopter h):

$$\forall h \in H: \sum_{f \in F_h} \sum_{s \in S} x_{f,h,s} \leq N_h$$

- iv. Landings after 5:00 PM are only allowed on platforms with a night-shift crew

$$\forall f \in C^{iv}, \forall h \in H_f, \forall s \in C_f^{iv}: x_{f,h,s} = 0$$

- v. 3.5 hours gap before reusing the same helicopter

$$\forall h \in H, \forall s \in S: \sum_{f \in F} \sum_{h \in C_s^v} x_{f,h,s} \leq 1$$

- vi. 3.5 hours gap before revising the same platform

$$\forall d \in C^{vi}, \forall f_1 \in C_d^{vi}, \forall f_2 \in C_{d,f_1}^{vi}, \forall s \in C_{d,f_1}^{vi}: \sum_{h \in H} x_{f_1,h,s} + \sum_{h \in H} \sum_{2s \in C_{d,f_1}^{vi}} x_{f_2,h,s} \leq 1$$

- vii. Binary decision variables

$$\forall f \in F, \forall h \in H_f, \forall s \in S: x_{f,h,s} \in \{0, 1\}$$

Details about the different sets and parameters used in the model can be found in the [appendix](#). The solution to this minimization problem is a flight schedule respecting the logistical constraints. The Python PuLP library with the COIN-OR Branch and Cut solver for linear programming models is used to solve this minimization problem (PuLP). The PAX algorithm also allows a set of constraints to be relaxed in case no solution to the initial formulation has been found. In fact, for a given set of flights to schedule, the PAX algorithm will solve the minimization problem with all the operational constraints (from (i) to (vii) included) activated. In case no solution is found, the PAX algorithm will start to relax certain constraints, in agreement with the operational teams, to ensure that a flight schedule respecting as many constraints as possible can be found. An example of a relaxed constraint is to gradually lower the 3.5 hours gap in constraints (v) and (vii) from 3.5 to 2 hours with, e.g., a 0.5-hour step.

Implementation

Our operational recommendations are sent daily to the aviation super intendent responsible for adjusting flight schedules over the next 14 days. Based on our recommendations, each daily helicopters' planning can be modified thanks to exhaustive reports.

Process

Every night, the optimized helicopters' routings are automatically calculated for each of the next 14 days and based on the latest reservations updates. Immediately afterwards, 14 horizons are calculated to summarize the operational recommendations. At 6AM, an email notification is sent to the helicopter booking team with a summary of the potential savings following the computed recommended routing and scheduling modifications for the 14 coming days. The email notification contains a link to an interactive web interface containing the exhaustive reports of these recommended modifications for each day.

Report

Each daily report contains several parts, including: a ranking of the scenarios and the actual planning with respect to different metrics, the actual planning of the report's day and for each scenario the recommendations summarized in two tables and one Sankey diagram. The first table is the optimized planning as proposed by the scenario. A Sankey diagram allows visualization of the migration of the passengers from the actual to the optimized flights, and the details of the rebooked passengers are found in the second table. [Fig. 3](#) presents an example of a scenario extracted from a daily report.

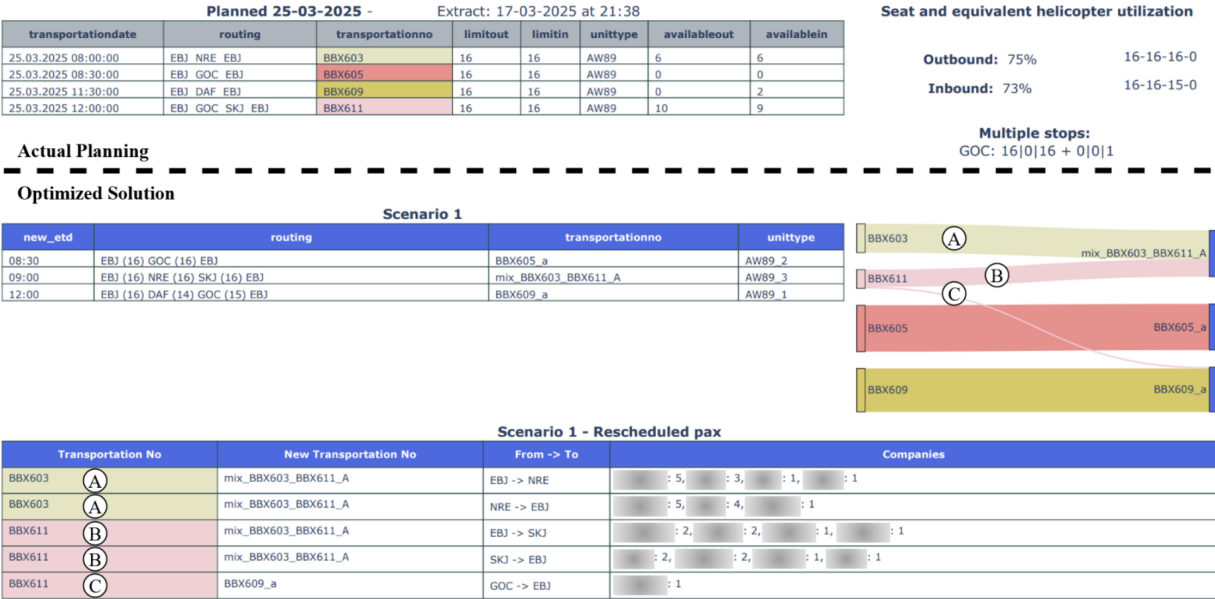


Figure 3—Extract of a report created in the evening of March 17th, 2025, presenting March 25th, 2025, actual planning and optimization results. Above: Actual planning (left), average and marginal seat utilization (right). Below: optimized daily routing and scheduling scenario composed of optimized planning, a Sankey diagram helps visualization of the flux of passengers (A, B, C) from the actual to the optimized solution flights, and the passengers' information involved (A:20, B: 12, C: 1).

Discussion

Our solution was introduced to the operations team in July 2024 and was already supporting the helicopters operations team with solid results. Our method is currently being extended to other use cases than optimizing the actual daily helicopters scheduling and planning. Indeed, the same method can be applied to simulate adding (break-in) or rebooking (break-out) reservations to another day, which in turn can reveal opportunities for further flight savings.

Marginal Flight Justification

The concept of marginal flights was introduced to have a better picture of the usage performance of our helicopters' actual plannings. Fig. 4 highlights the importance of marginal flight seat utilization instead of the average seat utilization.



Figure 4—Difference between average and marginal flights seat utilization: two cases with respectively 6 (5+1) and 5 (4+1) flights, having the same average seat utilization and a different marginal seat utilization.

The seat utilization corresponds to the sum of flying passengers, divided by the total number of seats. The average seat utilization does not picture correctly the number of empty seats with respect to the capacities of the helicopters. Instead, we used as a performance metric marginal flight seat utilization as a rendering of the flight saving opportunities.

The marginal flight is a fictive picture obtained as follows, separately for outbound and inbound flights: we group all the passengers of the actual planning in the flights ordered by decreasing capacities, until the very last flight, being the so-called marginal flight. It can be also seen as the smallest capacity flights filled with the total number of empty seats. Marginal flight seat utilization is the average seat utilization of the outbound and inbound marginal flight seat utilization.

Results

The implementation of the offshore platforms operated by TotalEnergies in the Danish North Sea is such that saving flights becomes the primary driver of cost reductions. On average, the distance between two platforms is 5 times smaller than their distance to Esbjerg airport, 10 times for both ways, making a flight saving almost 10 times more impactful than saving a touchdown.

In terms of the flying hours cost, the ideal theoretical route would be one flight containing all the necessary stops. Generally, the helicopters' capacities or range do not permit it. Indeed, the helicopters capacities maximum is about ten times smaller than the number of daily reservations; and offshore refueling is not always possible. Google vehicle routing optimization model cost function being mainly driven by the flying time, naturally favors the minimum flights with as many stops as possible, rather than multiple flights.

In addition to being an activity that includes risk, a touchdown requires offshore workers to secure the landing and takeoff, interrupting their activities for approximately half an hour and thereby impacting production. Therefore, efforts are made to minimize the number of visits to the same platform within a single day. For those reasons our solution has been finding optimized tradeoffs between cost optimization and operability.

Between July 2024 and April 2025, 21 flights were directly saved after the suggestions of our solution.

This reduction in flights resulted in approximately 48 flying hours i.e. 58 tons of CO₂ emissions avoided and \$525,000 in logistics cost savings. Each flight saved also contributed to lower operational risk and improved safety by reducing offshore landings.

Fig.5 represents the evolution of the daily number of flights and usage performance, since January 2023. A main project continuously impacted the number of flights until Summer 2024, and only sporadically until the end of year 2024. Our solution was integrated into the aviation operations team in the Summer 2024, some flights were still frozen, and the optimization limited. Fewer frozen flights allow more flexibility to our optimization models, widening the solution space, leading to more flight saving opportunities. Nonetheless, our solution misses saving flights if the daily number of empty seats accounts for a marginal flight usage of 1-4 reservations. These passengers could be rebooked to another day to save a flight.

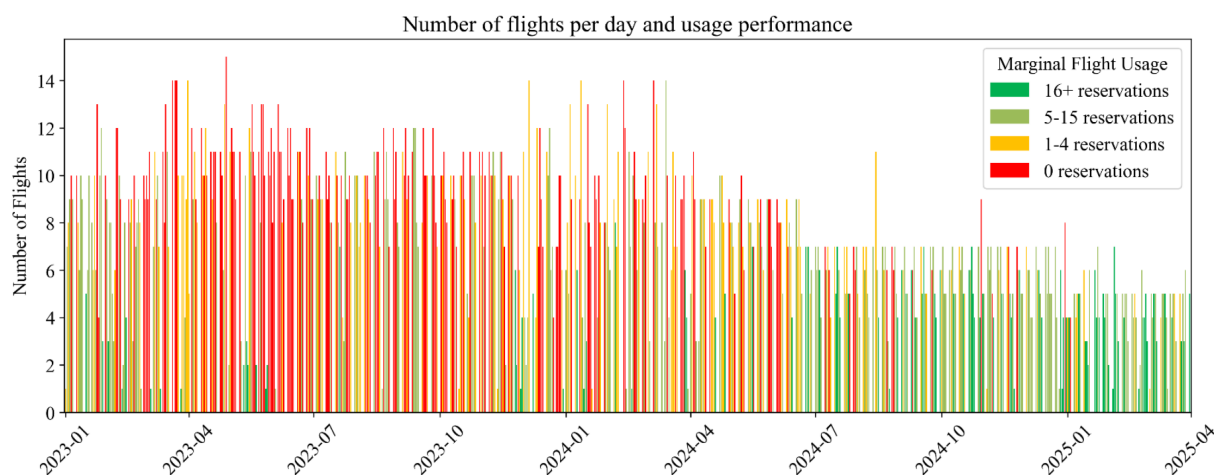


Figure 5—Evolution of daily number of helicopters flights from January 2023 to April 2025. Our solution was fully integrated in August 2024 after some tests were conducted in July 2024.

Vision / Current Developments

Our current developments further challenge the daily passengers' reservations. We discuss here the opportunities of modifying the daily reservations, by adding or rebooking helicopters' reservations, within the last 14 days before the flying date. Those modifications are known as 'break-in' and 'break-out' respectively.

The planning team needs to evaluate the impact of a 'break-in'. They need a decision support when no seat is available for the 'break-in', and the planning must optimally adapt to the new reservations. Using our solution's process to optimally add reservations to daily planning is straight forward. We simply run the process described in the previous sections but with a modified input including the 'break-in' reservations. For each simulated 'break-in', the optimization result is a report as described in the Method section.

A 'break-out', which involves agreeing to rebook a flexible passenger to a different date, is a more complex procedure but has significant savings potential, as mentioned previously. To that end, our solution has started to tackle the challenge of rebooking passengers only within the direct consecutive days, before and after their flying date.

The 'break-out' integration into our solution is as follows. First, we identify the days when the helicopters' usage performance of the marginal flights is between 1 to 4 reservations. Second, we identify the rebook able reservations for those dates, i.e. the people who are likely to be rebooked the flying day before or after. Several reservations are filtered out, if they cannot easily be rebooked the day before or the day after, day trippers, back-to-backs, etc. Third, we run a CP model to optimally select groups of rebook able people that are all leading to different input to the vehicle routing model. Once the groups are identified, they are canceled from the flying date planning, and the remaining reservations are used as input to run the same process as described in the previous sections, leading to a 'break-out' report.

Conclusion

The successful deployment of the PAX solution marks a significant achievement: we managed to translate advanced operational research from theory into tangible results within a real-world, complex offshore logistics environment. By applying customized optimization models and integrating Google OR-Tools into the affiliate daily operations, our team proved that operations research techniques can drive impactful efficiencies and cost savings.

Since its deployment in Summer 2024, PAX has delivered measurable improvements in operational cost-efficiency, sustainability through reduced CO₂ emissions, and enhanced safety by minimizing the number of flights and offshore touchdowns—key priorities for the energy sector. Typically, 21 flights saved between mid-2024 and mid-2025 represent 0.5M\$ of savings, and close to 60 tons of CO₂, without considering the route / distance optimizations.

The modular approach of PAX, encompassing split delivery, routing, seat allocation, and scheduling, ensures flexibility and adaptability to real-world constraints. Marginal flight analysis provides a more accurate metric for helicopter utilization, refining decision-making and supporting cost and safety objectives. The exploration of 'break-in' and 'break-out' scenarios further improves planning efficiency and unlocks additional opportunities for cost savings and risk reduction.

As the solution evolves, its scalability and adaptability to other affiliates and use cases is confirmed. By fostering a virtuous cycle of logistics excellence, cost-efficiency, and environmental responsibility, PAX directly supports ADIPEC's focus on safe, sustainable, and efficient operations. Future developments will enhance dynamic constraint integration and expand the solution's scope to address emerging challenges in offshore logistics, further strengthening its alignment with industry priorities.

References

- Vehicle Routing | OR-Tools. (n.d.). Google for Developers. Retrieved April 11, 2025, from <https://developers.google.com/optimization/routing>
- User Guides — PuLP 3.0.2 documentation. (n.d.). Retrieved April 11, 2025, from <https://coin-or.github.io/pulp/guides/index.html#>
- CP-SAT Solver | OR-Tools. (n.d.). *Google for Developers*. Retrieved April 11, 2025, from https://developers.google.com/optimization/cp/cp_solver
- Trefethen, F. (1954). A history of operations research. In *Operations research for management* (Vol. 1).
- Dror, M., Laporte, G., & Trudeau, P. (1994). Vehicle routing with split deliveries. *Discrete Applied Mathematics*, **50**(3), 239–254. [https://doi.org/10.1016/0166-218X\(92\)00172-1](https://doi.org/10.1016/0166-218X(92)00172-1)
- Archetti, C., Speranza, M. G., & Savelsbergh, M. W. P. (2008). An Optimization-Based Heuristic for the Split Delivery Vehicle Routing Problem. *Transportation Science*, **42**(1), 22–31. <https://doi.org/10.1287/trsc.1070.0204>
- Ahkamiraad, A., & Wang, Y. (2018). Capacitated and multiple cross-docked vehicle routing problem with pickup, delivery, and time windows. *Computers & Industrial Engineering*, **119**, 76–84. <https://doi.org/10.1016/j.cie.2018.03.007>

Appendix

Scheduling problem

Sets

- H : available helicopters
- F : flights to perform
- S : possible ETDs sorted from earliest to latest
- D : destinations to visit during the day
- H_f : available helicopters for flight f
- F_h : possible flights for helicopter h
- C_{iv} : flights and ETDs for constraint (IV)
- C_v : forbidden ETDs for constraint (V)
- C_{vi} : destinations, flights and forbidden ETDs for constraint (VI)

Parameters

- N_s : maximum number of flights for ETD s
- T_f : flight f time duration (in hours)
- C_h : seat capacity of helicopter h
- P_h : flying hour price for helicopter h (in DKK)
- P_{fuel} : helicopter fuel price (in DKK/l)
- N_h : maximum number of daily flights for helicopter h
- $Id(s, S)$: index of ETD s in set S
- K_h : hourly fuel consumption of helicopter h (in l/h)